

Determining Solutions to Inequalities

Teacher's Edition

Standard and Targets

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Learning targets I can determine which values from a set make an inequality true. I can determine whether a graph represents a solution set. I can solve two-step inequalities and represent the solutions on a number line.

Essential questions How do you determine whether a value makes an inequality true? How do you determine the solution to an inequality? How many solutions can an inequality have?

Building on MA.7.AR.2.1 — one-step inequalities in one variable, solutions represented algebraically or graphically.

Working toward MA.912.AR.2.6 — one-variable linear inequalities including compound inequalities.

Where this sits on the test This is the first lesson of the course on a **Grade 8 FAST benchmark**. Lessons 1 through 7 were all grade 7 codes carried by course 1205050 and not reported on the test. MA.8.AR.2.2 is scored, in the **Algebraic Reasoning** reporting category — 22–28% of the test, nine items. On the 2025 released items the Algebraic Reasoning benchmarks scored lowest of the four categories. Lessons 8, 9 and 10 are all this one benchmark, so this is three consecutive days on tested content for the first time all year.

No grade 8 bridge questions on this sheet. Worksheets 6 and 7 each ended with two MA.8.AR.2.1 items because those lessons sat on an untested grade 7 code. That is no longer needed — every question here is already at the tested benchmark.

Materials Whiteboards and markers. Calculators. Worksheet 7A.01.08. No cards to cut — the matching activity has been rebuilt (see below).

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: diagnostic question	2	4 min
Notes I–IV: what a solution is, the equal bar, the reversal rule, circles and shading	3–6	13 min
Example 1: is it a solution? no solving	7–9	4 min
Example 2: now solve it	10–12	5 min
Example 3: graph $x \geq -3$	13–15	4 min
Example 4: Talia and Jen — incomplete is not the same as wrong	16–18	5 min

Segment	Slides	Time
Whiteboards: 9 questions, each with its reveal	19–36	12 min
IXL — start today's skills	37	5 min
Total		53 min

Where the lesson lives: Notes III (slide 5) and Example 4 (slides 16–18). Notes III carries the one rule that separates inequalities from everything they have done since Lesson 3. Example 4 is the difference between an incomplete answer and a wrong one, which is a distinction the test rewards and students have never been asked to make.

Read this before you plan the day. Math Nation's components total 45 to 65 minutes — 5 for the Warm-Up, 10–15 for the Exploration, 15–20 for the Guided Practice, 5–10 for Your Turn!, 5–10 for the Collaboration and 5 for the Wrap-Up — before any worksheet exists. At the top of that range it is twelve minutes over a full period. The plan above fits because the Exploration becomes Example 1, the Guided Practice's side-by-side table becomes Example 2, and the Collaboration's card match becomes the whiteboard sequence.

No partner or group work anywhere in this lesson. Math Nation has five instances: the Exploration's questions 2 and 5 are “discuss with your partner,” Guided Practice 1b is partner discussion, 1.8.5 is a partner card-matching activity, the Clarify/Critique/Correct routine has partners alternating steps of a solve, and the Wrap-Up tip is to pair students. The card match is now part of the whiteboard sequence, where every student decides the circle and the direction on their own board.

Vocabulary

inequality — a statement that two expressions are not necessarily equal

solution set — every value that makes an inequality true

boundary value — the number where the two sides are exactly equal

open circle — the boundary is not a solution — used for $<$ and $>$

closed circle — the boundary is a solution — used for $\leq$ and $\geq$

A note on the glossary. Math Nation lists inequality as the only key term. Solution set, boundary value and the two circles are added because the lesson's whole argument is about which values belong and whether the edge is one of them, and students cannot discuss that without names for the parts.

Slide-by-Slide Teaching Notes

Slide 1. Determining Solutions to Inequalities *1 min*

- Post the learning target.
- This is the first grade 8 benchmark in the unit and it is on the FAST.
- Say the framing exactly as written: nothing they know is being replaced, one rule is being added, and it fires on exactly one kind of step.

Slide 2. Warm-Up 4 min

- TYPE: Diagnostic question.
- INDEPENDENT, boards up together.
- ANSWER: 2.
- Every distractor is a named misconception — 1 is ordering negatives by size, 3 is not looking, 4 is the belief that negatives cannot be compared.
- On a number line -4 is to the right of -6 , so it is greater.
- This exact comparison is the reason the reversal rule exists, and it arrives in Notes III in four minutes.
- OFF THE SLIDE, YOURS TO SAY: Boards up together, one number each. Then a sentence saying how you decided.

Slide 3. Notes: I. A solution is any value that makes it true 3 min

- The panel is the practical point and it is what whiteboard questions 1 to 3 use.
- Testing is not a lesser method — on a select-all item it is often the fastest one, and students should know they are allowed to.
- OFF THE SLIDE, YOURS TO SAY: You can test a value without solving anything. Substitute, and see.

Slide 4. Notes: II. The equal bar decides whether the boundary counts 3 min

- Small and worth its own slide, because whiteboard questions 2 and 3 differ by exactly this bar and nothing else.
- Read the four symbols out loud; 'at most' and 'at least' are the phrasings the FAST uses and they are not obvious.
- OFF THE SLIDE, YOURS TO SAY: One bar under the symbol, and one number moves from outside the answer to inside it.

Slide 5. Notes: III. Multiplying or dividing by a negative reverses it 4 min

- THE NEW RULE, and item D is the warm-up answering it.
- Do not present the reversal as a convention to memorize — the warm-up has just established that $-4 > -6$, and the rule is only that fact written generally.
- Students who saw the warm-up and believe item D will not need to memorize anything.
- OFF THE SLIDE, YOURS TO SAY: The reversal happens once, at that step. Not at the end, and not twice.

Slide 6. Notes: IV. Open circle, closed circle, and which way to shade 3 min

- Item C is where a graphed question is decided.
- The solution and its graph are one point together, so a student who solves correctly, reverses correctly, and then shades from the symbol they were given loses the whole point having done all the mathematics.
- Say it, and then say it again when you graph in Example 3.
- OFF THE SLIDE, YOURS TO SAY: Once you have reversed, the original inequality points the wrong way.

Slide 7. Example 1 1 of the 4 min

- Whiteboards.

- Substituting, not solving — this is Notes I.
- Both take ten seconds and the second one is the interesting one.

Slide 8. Example 1 *2 of the 4 min*

- The $x = 3$ column is the whole example. 7 is not less than 7, so 3 is not a solution — and it is the boundary, which is exactly the value students assume must be included.
- Notes II.

Slide 9. Example 1 *1 of the 4 min*

- Notes II demonstrated rather than asserted.
- This is whiteboard questions 2 and 3, so the round will move fast if this slide lands.
- OFF THE SLIDE, YOURS TO SAY: One bar. Same number, opposite verdict.

Slide 10. Example 2 *1 of the 5 min*

- SPOKEN ONLY.
- No — we add 5 and then divide by positive 4, and neither of those reverses.
- Asking the question before solving is the habit; the answer being 'no' this time is what makes the answer being 'yes' next time mean something.

Slide 11. Example 2 *2 of the 5 min*

- Name the operation at each line and say 'no reversal' out loud both times.
- The panel closes the loop with Example 1 — testing and solving agree, which is worth pointing at because students often believe they are different topics.
- OFF THE SLIDE, YOURS TO SAY: And $x = 3$ is a solution, which is what Example 1 found by substituting.

Slide 12. Example 2 *2 of the 5 min*

- Put this beside the previous slide.
- The first two lines are identical in structure; only the last one differs, and it differs because of a minus sign that was there from the beginning.
- Have them test $x = 0$ in the original: $-5 \leq 7$ is true, and $0 \geq -3$, so the answer agrees.
- OFF THE SLIDE, YOURS TO SAY: One sign changed, and the answer is now everything on the other side of the line.

Slide 13. Example 3 *1 of the 4 min*

- SPOKEN ONLY.
- Closed, because of the equal bar; shaded right, because x is greater than -3 .
- Both decisions come from the SOLVED line, which is Notes IV item C.

Slide 14. Example 3 *2 of the 4 min*

- Draw it on the board as they draw it on paper.
- Point at the circle and say why it is filled, then at the arrow and say why it points right.
- Two decisions, two reasons, every time.

Slide 15. Example 3 *1 of the 4 min*

- Testing one value from each side is the check that catches a shading error, and it is the same substituting from Notes I.
- Students who cannot remember which way to shade can always find out this way.
- OFF THE SLIDE, YOURS TO SAY: One number from each side of the circle. That is the whole check, and it takes twenty seconds.

Slide 16. Example 4 *2 of the 5 min*

- Whiteboards, the second column only — substitute each value into $8 - 1.2c \geq 5.6$ and say yes or no.
- Offering “neither” as a live option is the point of the question; Math Nation’s version forces a choice between two wrong answers.
- OFF THE SLIDE, YOURS TO SAY: Substitute. Do not solve. One of them may be right, or neither.

Slide 17. Example 4 *2 of the 5 min*

- Take the three rows in order and let the pattern arrive on its own.
- Talia’s answers are never false — she just misses $c = 0$.
- Jen’s include $c = 5$, which is not a solution at all.
- Do not name the difference yet; the next slide is where it gets its two words.
- OFF THE SLIDE, YOURS TO SAY: Neither one is right, and they are not wrong in the same way.

Slide 18. Example 4 *1 of the 5 min*

- Dividing by -1.2 reverses the symbol, so the answer is $c \leq 2$ and neither student has it.
- Give the two mistakes their names out loud — incomplete, and incorrect — because the test rewards the distinction and students have never been asked to make it.
- Worksheet question 6 is the same shape — two students, neither right — though there both answers are simply incorrect rather than one being incomplete.
- OFF THE SLIDE, YOURS TO SAY: Talia named only some of the solutions. Jen named numbers that are not solutions at all.

Slide 19. Whiteboards • Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 20. Whiteboards • Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline: substitute, do not solve.
- Everything after this changes exactly one thing.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 21. Whiteboards · Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 22. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the value tested.
- It is the boundary, and with a strict $<$ the boundary is outside.
- Expect yes from anyone who computed 7 and stopped reading.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 23. Whiteboards · Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 24. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT EARNS ITS PLACE.
- One feature changed from question 2: the equal bar.
- Same value, same arithmetic, opposite verdict.
- Notes II is one small rule and this pair is the whole of it.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 25. Whiteboards · Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 26. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: solve rather than test.
- Nothing reverses — adding does not, and dividing by a positive does not.
- And $x = 3$ is included, which agrees with question 3.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 27. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 28. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THE LESSON EXISTS FOR.
- One feature changed: the sign of the coefficient.
- The first step is identical; the second divides by a negative and the symbol turns over.
- A board saying $x \leq -3$ did the arithmetic and not the rule.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 29. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 30. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSSES IT.
- One feature changed: draw it rather than write it.
- Closed because of the bar, right because the SOLVED line says greater — not because the original said $\leq$.
- Ask which of the two inequalities on question 5 they read to decide the direction; the answer must be the second one.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 31. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 32. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: five values at once instead of one, which is how the FAST asks it.
- Solve first — $x > -5$ — and then reading off the list takes seconds.
- THE TWO BOARDS TO LOOK FOR: one leaves out -4.999 , because it LOOKS like -5 and -5 is out; and one puts -5 IN, which is questions 2 and 3 of this round arriving again, because the symbol is strict and the boundary is outside. -4.999 is greater than -5 and it is in by a thousandth.
- Substituting it gives -24.996 , which is greater than -25 and settles the argument in one line.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 33. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 34. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: a fraction coefficient.
- NOTHING REVERSES — $3/2$ is positive, and it is worth saying out loud that a fraction is not a negative.
- THE BOARD TO LOOK FOR SAYS $b \leq 6$, from a student who has learned "fractions are strange" and turned the symbol over for safety.
- The rule is one thing only: multiply or divide by a NEGATIVE.
- Ask them which number here is negative.
- None of them is.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 35. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 36. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: NEITHER.
- Add 3 to get $-y/7 < -2$, multiply by -7 and reverse: $y > 14$.
- Not -14 .
- TWO SEPARATE ERRORS ARE ON OFFER HERE and the question is built so that picking a side means missing one of them: Jonah got the sign of the boundary right and the direction wrong, Tanner got the direction right and the sign wrong, and a student who "agrees with Tanner" has been rewarded for half an answer.
- Substitution settles it without any algebra at all — $y = 0$ gives -3 , which is not less than -5 , so Tanner is out; $y = -21$ gives 0 , which is not less than -5 , so Jonah is out; $y = 21$ gives -6 , which is.
- Being offered two answers is not evidence that one of them is right, and that is the habit this question is really for.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 37. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.

- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.08 — Answer Key

1. $-25 < 4x - 5$	$x > -5$: -4.999, $-1\frac{6}{17}$, 0, $6\frac{3}{8}$, 12.75, 42 Six of the ten. -4.999 is a solution and -5 is not; the difference is one thousandth, which is why they sit in the same table.
2. $-25 > 4x - 5$	$x < -5$: -20.6, -10.5, -7.25 Three of the ten. -5 appears in neither list.
3. rewrite for -5	$-25 \leq 4x - 5$ and $-25 \geq 4x - 5$ $4(-5) - 5 = -25$ exactly, so -5 is the boundary. The equal bar is what lets a boundary value count.
4. how many solutions	<i>infinitely many</i>
5. negative multiplier	<i>reverses</i> Accept “flips,” “switches direction,” or the symbol drawn the other way round.
6. Tanner and Jonah	<i>Neither is correct. $y > 14$</i> $-y/7 < -2$, then multiply by -7 and reverse: $y > 14$. Both students carried the negative into the boundary; $(-2)(-7) = +14$. Jonah also failed to reverse the symbol.
7. $5 + \frac{2}{3}b \geq 9$	$b \geq 6$ closed circle at 6, shade right $\frac{2}{3}b \geq 4$, then multiply by $3/2$. No reversal.
8. $4.6c - 9.5 < 4.3$	$c < 3$ open circle at 3, shade left $4.6c < 13.8$. Dividing by a positive leaves the symbol alone.
9. $-1 \leq -4 - y/6$	$y \leq -18$ closed circle at -18, shade left $3 \leq -y/6$, then multiply by -6 and reverse to get $-18 \geq y$. Rewrite as $y \leq -18$ BEFORE graphing — this is the item that gets shaded backwards.
10. $-7x - 12 > 16$	$x < -4$ open circle at -4, shade left $-7x > 28$, then divide by -7 and reverse. Checking $x = -5$ (true) and $x = 0$ (false) confirms the direction.

Question 6 is the one Math Nation gets wrong. Their homework poses this exact structure — two students, two answers — and asks “Who do you agree with?” with no third option. In their version the inequality is $8 - 1.2c \geq 5.6$, the answer is $c \leq 2$, and neither Talia nor Jen has it. Their own Practice sheet asks the same question with the words “or do you agree with neither,” so the fix already exists in their materials. Ours uses their Practice wording throughout.

Question 9 is the shading trap. Solving gives $-18 \geq y$, and a student who graphs straight from that line shades right. Rewriting as $y \leq -18$ first is the habit; it is Notes IV item C — shade from the SOLVED line — and it costs nothing.

Number lines are printed for every graphing item. Each one is scaled and labeled around the correct boundary, so the only two decisions left are the circle and the direction — which are the two things being assessed. Math Nation’s Lesson 10 prints unlabeled lines, which cannot be graded; their own teacher tip suggests supplying numbered ones.

Additional Practice 7A.01.08 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. $-3 + x/2 < -7$	$x < -8$: $-25, -17.3, -11.5, -9.5$
2. $-3 + x/2 > -7$	$x > -8$: $-7.75, -0.1212, 6\frac{2}{7}, 6\frac{3}{8}, 36$ -8 is in neither list; at $x = -8$ the two sides are exactly equal.
3. rewrite for -8	$-3 + x/2 \leq -7$ and $-3 + x/2 \geq -7$
4. how many solutions	<i>infinitely many</i>
5. Marco and Elena	<i>Elena is correct. $x < -8$</i> -0.5x > 4, then divide by -0.5 and reverse. Marco divided without reversing. Unlike the worksheet's question 6, one of these two students is right — students who have learned “the answer is always neither” get caught here on purpose.
6. negative multiplier	<i>reverses</i>
7. $-8 \leq a/(-3) - 9$	$a \leq -3$ closed circle at -3, shade left $a/(-3)$ is the same as $-a/3$. A negative denominator reverses the symbol exactly as a negative coefficient does.
8. $9m + 16 > -20$	$m > -4$ open circle at -4, shade right
9. $2.8 < -1.3p - 0.5$	$p < -33/13$, about -2.54 open circle between -3 and -2, shade left The boundary is not a whole number. Place the circle a little past halfway from -2 toward -3 and say in words which two integers it lies between. Estimating a boundary onto a labeled line is a FAST skill in its own right.
10. $-2 + (5/4)x \geq -12$	$x \geq -8$ closed circle at -8, shade right Multiplying by the reciprocal $4/5$ is positive, so the symbol stays put.

Misconceptions to Watch

- **Not understanding why the symbol reverses for a negative multiplier.** Named first in Math Nation's list. Notes III item D and the whiteboard sequence. (Math Nation TE)
- **Not knowing when to use an open circle and when to use a closed one.** Also named. Notes IV items A and B, and the whiteboard sequence. (Math Nation TE)
- **Shading toward the direction the symbol appears to point.** $x > -5$ shaded left. Notes IV item D — test a value instead.
- **Graphing from an answer written with the letter on the right.** $-18 \geq y$ shaded rightward. Worksheet question 9.
- **Reversing the symbol at the end of every problem once they have seen one reversal.** The Notes III panel — the reversal happens once, at that step, not at the end and not twice — and the whiteboard sequence, where the reversal arrives after four questions in which nothing reverses.
- **Treating the boundary value as a solution when the symbol is strict.** Example 1 and whiteboard questions 2 and 3.

- **Judging an answer “wrong” without distinguishing incomplete from incorrect.**
Slides 16–18 and worksheet question 6.

Changes from the Math Nation Lesson

- The error-analysis question was repaired. Math Nation's homework question 5 asks students to choose between Talia's $c \leq -2$ and Jen's $c \geq -2$ for $8 - 1.2c \geq 5.6$. The correct answer is $c \leq 2$ and neither student has it, but the question offers no third option. Their own Practice sheet asks the same kind of question with “or do you agree with neither,” which is the wording used here. The item became slides 16–18 and the sharpest five minutes in the lesson.
- The warm-up was replaced. Math Nation's 1.8.1 is a 2:24 career-profile video about foresters followed by two “answers will vary” reflection questions — no mathematics at all, on the first day of a tested grade 8 benchmark. The new warm-up is a diagnostic question on -4 against -6 , which is the single fact the reversal rule is built from.
- A teacher tip that described a different problem was dropped. The printed tip for Guided Practice question 3a discusses supporting students through “ $-a/8$ ” and writes the equivalence $-(a/8) = (-a)/8 = a/(-8)$ three times. The item on the page is $2 - a/6 \geq -4$.
- Additional Practice question 9 keeps its awkward boundary on purpose. $2.8 < -1.3p - 0.5$ gives $p < -33/13$, about -2.54 , which does not land on a tick mark. Rather than retuning the numbers, the key asks students to place the circle between -3 and -2 and name the two integers in words. Estimating a boundary onto a labeled line is a FAST skill and this is one of only three items on the lesson WORKSHEETS that requires ESTIMATING where the boundary falls between two ticks (the unit assessment and the practice test each carry two more, at questions 9 and 10) (Lesson 10's question 6 also lands off a tick, at exactly 6.5 , but a midpoint is placed rather than estimated) — the others are worksheet question 5 and Additional Practice question 4 in Lesson 10.
- Additional Practice question 5 was rewritten so that one student IS right. After the worksheet's “neither” item, students who have learned the pattern rather than the mathematics will answer “neither” again; Elena's $x < -8$ is correct and catches them.
- Number lines are printed under every graphing item, scaled and labeled around the correct boundary. Math Nation supplies unlabeled lines in Lesson 10 and none at all on the homework.
- Every partner and group activity is replaced — five of them. The card-matching Collaboration and the Clarify/Critique/Correct alternating-steps routine are both in the whiteboard sequence; the Exploration's partner discussions are Example 1 worked as a class.
- The word “homework” was removed. Math Nation's tip reads “if time is an issue, consider assigning 1.8.4 Your Turn! for homework.” Worksheet 7A.01.08 is classwork; the Additional Practice goes to Google Classroom for students who want more.
- Grammar was corrected in two printed answers: “the answers are graph differently” and “an inequality can be graph with an open or closed circle.”
- The Wrap-Up was replaced with a Ticket Out the Door that produces gradeable evidence: a substitution check on a boundary value, a solve that reverses, and a graph the student has to label.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- Nothing in the mathematics was changed.